Database Fundamentals

1. **Data vs. Information:**
   a) Discuss the key differences between data and information.
   b) Explain how data is transformed into information.
   c) Provide examples of raw data and how it can be processed into meaningful information.
2. **Information Management:**
   a) Describe the importance of information management in today's world.
   b) Outline the key challenges associated with managing large volumes of information.
   c) Explain how effective information management can benefit organizations.
3. **Databases and Database Management Systems (DBMS):**
   a) Define the concept of a database and its core components.
   b) Explain the role of a Database Management System (DBMS) in managing databases.
   c) Discuss the advantages of using a DBMS compared to traditional file systems for data storage and retrieval.
4. **Database Applications:**
   a) Describe how applications interact with databases.
   b) Provide examples of different types of applications that rely on databases (e.g., e-commerce, library management systems).
   c) Explain how database design principles can influence the efficiency and effectiveness of an application.
5. **Choose one of the following topics and discuss it in detail:**
   a) **Data Integrity:** Explain the importance of data integrity in a database and discuss various methods to ensure data accuracy and consistency.
   b) **Database Security:** Describe potential security threats to databases and outline strategies to protect sensitive information.
   c) **The Relational Model:** Explain the basic principles of the relational database model, including tables, columns, rows, and relationships between them.
6. Compare and contrast the two-tier and three-tier database architecture models. Discuss the advantages and disadvantages of each model and explain when you might choose one over the other. Consider factors such as scalability, performance, and security.
7. In the context of modern web applications, evaluate the importance of the n-tier architecture (where $n \geq 3$). Explain how the separation of concerns

between presentation, business logic, and data access layers contributes to maintainability, security, and efficient resource utilization.

8. Discuss the potential challenges associated with implementing a multi-tier database architecture. How can these challenges be addressed? Explore topics like data consistency, network latency, and deployment complexity.
9. Emerging technologies like cloud computing and mobile applications introduce new considerations for database architecture. How can the n-tier architecture be adapted to accommodate these trends? Discuss potential benefits and drawbacks of cloud-based database solutions for multi-tier applications.
10. Security is a critical aspect of any database system. Explain how the separation of concerns in a multi-tier architecture can contribute to enhanced data security. Discuss security measures that can be implemented at each tier (presentation, business logic, data access) to protect sensitive information.
11. Discuss the key differences between data and information.
12. Define the term information management.
13. How do file-processing systems and DBMS differ in their handling of:
    a) Relationships between data elements?
    b) Data integrity and consistency?
    c) Concurrency control for multiple users accessing the data?
14. Briefly explain the following: Relational database model and beyond relational database model.
15. How do ACID properties ensure data integrity in transactional databases? (Pick any two to explain with real-world examples).
16. What are the different types of database languages and how do they contribute to managing data within a database system?
17. Explain any three (3) core components of a database system.
18. Define an Entity-Relationship Model (ER Model) and explore the fundamental components that make it up, such as entities, attributes, and relationships.
19. Normalization is a crucial process in database design. What are three key advantages gained by normalizing a database?
20. Distinguish between primary and foreign keys in relational databases.
21. Differentiate between two-tier and three-tier database architecture models. Enumerate two advantages and one disadvantage of each tier database architecture.
22. Given the following attributes for a book database table (BookID, Title, Author, ISBN, Publisher, Year), write an SQL CREATE TABLE statement that defines the table structure, including data types and constraints. Assume

you want to ensure unique identification for each book and prevent missing titles or author names.

23. Assuming you have the following book information (Title: 'To Kill a Mockingbird', Author: 'Harper Lee', ISBN: '978-0446310727', Publisher: 'Penguin Random House', Year: 1960) and another book with the title 'The Lord of the Rings', by J.R.R. Tolkien, ISBN: '978-0547928225', published by Houghton Mifflin Harcourt in 1954, write an SQL INSERT query to populate the 'Book' table with these two entries.
24. ROADWAY TRAVELS "Roadway Travels" is in business since 1977 with several buses connecting different places in India. Its main office is in Hyderabad. The company wants to computerize its operations in the following areas: Reservations Ticketing Cancellations Reservations: Reservations are handled directly by the booking office. Reservations can be made 60 days in advance in either cash or credit. In case the ticket is not available, a wait listed ticket is issued to the customer. This ticket is confirmed against the cancellation. Cancellation and modification: Cancellations are also directly handed at the booking office. Cancellation charges will be charged. Wait listed tickets that do not get confirmed are fully refunded. AIM: Analyze the problem and come up with the entities in it. Identify what data must be persisted in the databases.
25. Consider the given tables in fig. 1, 2 and 3 below:

Figure 1: An Instance of **Sailors**

sid	sname	rating	age
22	Dustin	7	45.0
29	Brutus	1	33.0
31	Lubber	8	55.5
32	Andy	8	25.5
58	Rusty	10	35.0
64	Horatio	7	35.0
71	Zorba	10	16.0
74	Horatio	9	35.0
85	Art	3	25.5
95	Bob	3	63.5

Figure 2: An Instance of **Reserves**

sid	bid	day
22	101	10/10/1998
22	102	10/10/1998
22	103	10/8/1998
22	104	10/7/1998
31	102	11/10/1998
31	103	11/6/1998
31	104	11/12/1998
64	101	9/5/1998
64	102	9/8/1998
74	103	9/8/1998

Figure 3: An Instance of **Boats**

bid	bname	color
101	Interlake	blue
102	Interlak	red
103	Clipper	green
104	Marine	red

Write the sql statements to:

a) Find the names of sailors who have reserved boat number 103.
b) Find the names of sailors who have reserved at least one boat.
c) Find the names of sailors who have reserved a red or a green boat.
d) Find the **sid**s of all sailors who have reserved red boats but not green boats.
e) Find all sids of sailors who have a rating of 10 or have reserved boat 104.
f) Find the ages of sailors whose name begins and ends with B and has at least
g) three characters.
h) Compute increments for the ratings of persons who have sailed two different
i) boats on the same day.
j) Find the colors of boats reserved by Lubber.
k) Find all sids of sailors who have a rating of 10 or have reserved boat 104.
l) Find the names of sailors who have not reserved a red boat.
m) Find sailors whose rating is better than some sailor called Horatio.
n) Find the sailors with the highest rating.